Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2343611	Mini-Project - Full Stack Java Programming	–	2*+2	–	–	–	2	2

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II					
2343611	Mini-Project - Full Stack Java Programming	--	--	--	--	--	50	25	75

Lab Objectives:

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Sr. No.	Lab Objectives
The Lab experiments aims:	
1	To set up development environments for Java full-stack projects.
2	To develop a web interface using front-end technologies.
3	To build RESTful web services using Spring Boot.
4	To implement database CRUD operations using Hibernate/JPA.
5	To integrate front-end and back-end applications.
6	To deploy the application and present the capstone project.

Lab Outcomes:

Sr. No.	Lab Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy
On successful completion, of course, the learner/student will be able to:		
1	Setup and configure full-stack Java development environment.	L1, L2
2	Develop responsive user interfaces with HTML, CSS, and React.js.	L1, L2, L3
3	Create RESTful APIs using Spring Boot framework.	L1, L2, L3
4	Perform CRUD operations with MySQL database integration.	L1, L2, L3, L4
5	Integrate front-end and back-end with secure API endpoints.	L1, L2, L3, L4
6	Deploy full-stack application on cloud platform and present capstone.	L1, L2, L3, L4, L5, L6

Prerequisite: C /Python programming. Basic commands of Windows and Linux operating systems.

Hardware & Software Requirements:

Hardware Requirement:	**Software requirement:**
PC i3 processor and above. 8GB RAM (minimum), 16GB recommended, 256GB SSD recommended, Stable internet connection.	JDK 17+, IntelliJ IDEA / Eclipse, VS Code, MySQL / MongoDB, Postman, Git, GitHub, Spring Boot, Angular CLI., Node.js & npm (for frontend React.js)AWS / Heroku Account for Deployment

DETAILED SYLLABUS:

Sr. No.	Module	Detailed Content	Hours	LO Mapping
I	**Introduction to Full Stack Java**	Overview of full-stack development: Definition of Full Stack Development, Components of Full Stack: Frontend, Backend, Database, DevOps, Full Stack Developer Roles & Skills, Technology Stack Overview: Frontend: HTML, CSS, JavaScript, React.js, Backend: Java, Spring Boot, Database: MySQL, Tools: Git, Docker, Postman, Real-world applications of full stack, Java SE concepts: JVM, JDK, JRE, Data types, Variables, Operators, Control Statements, Introduction to OOP- Basic of OOP, Packages and Import statements, Exception Handling Basics, Introduction to Java EE- Need for Java EE in enterprise applications, Multi-tier architecture, MVC Architecture- Model: Business logic, Data management, View: User interface, Controller: Request handling, linking model and view. **Self-Learning Topics:** Application Servers vs. Web Servers, Java EE vs. Spring Boot: Modern development trends. MVC in Java Web Development.	**04**	**LO1**
II	**HTML 5, CSS 3 with Bootstrap**	**HTML Basics & Structure**: Learn HTML5 structure, semantic elements, character effects, document spacing, and working with tables, lists, and hyperlinks. **Responsive Web Design**: Learn to build websites that adapt to different screen sizes and devices using media queries, flexible layouts, and mobile-first design principles. **Image Handling & Forms**: Understand image roles, adding images to web pages, using images as links, and handling user input through various form elements (text fields, checkboxes, radio buttons, etc.). **CSS3 Fundamentals**: Explore CSS syntax, selectors, text formatting, fonts, colors, borders, and advanced features like CSS Grid. **Bootstrap Introduction & Components**: Get started with Bootstrap, utilizing grid systems and essential components like headers, dropdowns, and navigation bars.React.js basics, Fetch API, Axios for API calls. **Self-Learning Topics:** JavaScript DOM Manipulation, CSS Flexbox, Version Control with Git.	**04**	**LO2**
III	**Back-end development with Java**	Spring Boot introduction- What is Spring Framework? Why Spring Boot? Features of Spring Boot: Auto-configuration, Starter dependencies, embedded server, Spring Initializr, Project structure overview: pom.xml, application. Properties, Running your first Spring Boot application, RESTful API development- What is REST? REST vs. SOAP,HTTP Methods: GET, POST, PUT, DELETE, RESTful API Design principles, Creating REST Controllers in Spring Boot, Mapping HTTP methods to controller methods, Understanding Request Mapping, Get Mapping, Post Mapping, etc., Controllers and Services- Role of Controller in MVC, Defining Controllers in Spring Boot, @RestController vs. @Controller, Service layer responsibilities, Creating Service classes and injecting them	**05**	**LO3**

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		using @Autowired, Dependency Injection concepts, Repositories- Introduction to Spring Data JPA, Configuring database connection in Spring Boot (H2/MySQL), Entity classes and @Entity annotation, JpaRepository and CrudRepository interfaces, Basic CRUD operations: save(), findById(), findAll(), deleteById(), Exception Handling- Importance of exception handling, @ExceptionHandler annotation, Creating global exception handlers with @ControllerAdvice, Custom exception classes, Standard error responses (HTTP status codes, error messages).. **Self-Learning Topics:** Advanced Techniques of REST API and JPA.		
IV	**Database Integration**	Introduction to MySQL- What is a relational database? Introduction to RDBMS, Basics of MySQL: Database, Tables, Records, Primary & Foreign Keys, MySQL Workbench or command-line interface, Database design basics: ER diagrams, normalization, Setting up MySQL: installation, creating databases, tables, Introduction to SQL: Data Definition Language (DDL): CREATE, ALTER, DROP, Data Manipulation Language (DML): INSERT, UPDATE, DELETE, SELECT, CRUD operations- What is CRUD: Create, Read, Update, Delete, Writing SQL queries for CRUD operations: INSERT INTO table, SELECT FROM table, UPDATE table SET, DELETE FROM table, Filtering and Sorting data: WHERE, ORDER BY, LIKE, LIMIT, Using Aggregate Functions: COUNT, AVG, SUM, MAX, MIN, Joins: INNER JOIN, LEFT JOIN, JPA & Hibernate ORM, Repository interfaces, Query methods- Introduction to JpaRepository and Crud Repository interfaces, Default CRUD methods: save(), findById(), findAll(), deleteById(), Creating custom query methods using method naming conventions: findByName(), findByAgeGreaterThan(), etc.Pagination and Sorting with Spring Data JPA, Introduction to @Query annotation for custom JPQL/SQL queries, Native queries vs. JPQL.. **Self-Learning Topics:** JPQL Advanced Queries, Transactions in JPA, Optimizing JPA Performance.	**04**	**LO4**
V	**Full Stack Integration**	Connecting React frontend with Spring Boot backend- Overview of frontend-backend communication in full-stack applications, Setting up React app (Vite/CRA), Axios or Fetch API for making HTTP requests, Configuring CORS in Spring Boot for cross-origin access, Calling Spring Boot REST API from React components, Handling API responses in React (state management with useState/useEffect), API testing (Postman)- Introduction to Postman and its features, Creating API collections in Postman, Sending GET, POST, PUT, DELETE requests, Testing APIs with different payloads, Setting up environment variables in Postman, Automating tests with Postman scripts (basics), Error handling- Backend: Handling exceptions using @ControllerAdvice and @ExceptionHandler, Sending structured error responses (HTTP status codes, error messages, timestamps), Frontend: Handling API errors in React using try-catch and Axios interceptors, Displaying error messages to users in the React UI, Logging errors for debugging and maintenance, JWT for authentication- What is JWT? Structure of JWT: Header, Payload, Signature, Stateless authentication vs. session-based, Generating JWTs in Spring Boot, Validating and parsing JWTs, Security configuration using Spring Security, Storing JWT securely (localStorage/sessionStorage), Securing Frontend & Backend with JWT- Adding Authorization header in API requests from React, Protecting backend endpoints using JWT authorization, Role-based access control (RBAC) basics, Frontend: Securing routes and redirecting unauthenticated users, Refresh token strategy (basic	**05**	**LO4, LO5**

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		concept). **Self-Learning Topics:** Students can secure full-stack applications with JWT, protect API endpoints, and manage authentication flow.		
VI	**Deployment & Capstone Project**	Docker basics- What is Docker? Containers vs. Virtual Machines, Docker architecture: Images, Containers, Docker Daemon, Docker Hub, Setting up Docker environment, Writing Dockerfile: Basic syntax, Creating images for Java and Node.js applications, Docker commands:`docker build`, `docker run`, `docker ps`, `docker stop`, `docker rm`, `docker exec`, Managing Docker images and containers, Overview of docker-compose (introductory level), Deploying Spring Boot application- Creating Dockerfile for Spring Boot, Multi-stage builds for optimized Docker images, Exposing application port using `EXPOSE`, Environment variables and configuration management, Building and running Spring Boot Docker container, Accessing REST API from a containerized Spring Boot app, Hosting React application- Building React app for production (`npm run build`), Serving static files using NGINX or simple Node.js server, Writing Dockerfile for React application, Connecting React container to Spring Boot container, Basics of Docker networking (`docker network create`), Running both containers simultaneously using docker-compose, Capstone project implementation & presentation- Finalizing capstone project scope: features, architecture, roles, Implementing core functionalities (CRUD operations, authentication, API integration), Containerizing and deploying complete full-stack application, Preparing project documentation and presentation slides, Presenting project flow: architecture diagram, tech stack, demo. **Self-Learning Topics:** Full deployment with advanced techniques.	**04**	**LO6**

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Textbooks:

1. "Spring in Action" by Craig Walls
2. "Pro Spring Boot 3" by Felipe Gutierrez
3. "React Up & Running" by Stoyan Stefanov
4. "Java: The Complete Reference" by Herbert Schildt

Reference Books:

1. "Full Stack Development with Spring Boot and React" by Juha Hinkula
2. "Building Java Programs" by Stuart Reges, Marty Stepp

Online Resources:

Sr. No.	Website Name
1.	Spring Boot Official Documentation — https://spring.io/projects/spring-boot
2.	React.js Official Documentation — https://react.dev/
3.	FreeCodeCamp Full Stack JavaScript Tutorials — https://www.freecodecamp.org/
4.	Docker Documentation — https://docs.docker.com/
5.	MySQL Documentation — https://dev.mysql.com/doc/
6.	Postman Learning Center — https://learning.postman.com/
7.	Spring Initializr (https://start.spring.io/)

List of Experiments.

Sr No	List of Experiments	Hrs
01	Setup Java, Spring Boot, and React development environment.	01
02	Develop a static web page using HTML, CSS, and Bootstrap.	02
03	Build a dynamic front-end using React.js with API calls.	04
04	Develop a RESTful web service using Spring Boot.	02
05	Integrate MySQL database with Spring Boot application.	02
06	Implement JWT authentication in a Spring Boot application.	04
07	Connect React front-end with Spring Boot backend APIs.	04
08	Deploy the application using Docker and GitHub.	06
Sr No	**List of Assignments**	**Hrs**
01	Create a personal portfolio website using HTML, CSS, and React.	02
02	Develop a basic CRUD API using Spring Boot.	02
03	Implement a login and registration system with JWT.	02
04	Perform database operations using JPA Repository.	02
05	Build a single-page application (SPA) with React.js.	02
06	Deploy an integrated full-stack application on cloud/GitHub Pages.	02

Sr No	Capstone Mini- Project List (Sample Ideas)
01	**E-commerce Web Application** — Complete shopping system with admin dashboard.
02	**Online Library Management System** — For managing books, users, and borrowing transactions.
03	**Job Portal Application** — For job seekers and recruiters.
04	**Online Food Ordering System** — With user authentication, menu, and payment gateway.
05	**Student Course Registration System** — For colleges/universities.
06	**Event Management System** — Event creation, management, ticket booking.

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Note:- Capstone Mini-Project Guidelines.

1. **Objective of the Capstone Mini-Project:-** The capstone Mini-Project aims to enable students to design, develop, deploy, and present a real-world full-stack application using the Java technology stack, applying concepts of frontend, backend, database management, RESTful APIs, containerization (Docker), security (JWT), and deployment practices.

Students will work in a group of 4 students teams to build a functional application that solves a specific problem or fulfills a business need.

2. **Project Scope & Expectations**:- **Project Scale:** Should involve at least 2–3 core modules (e.g., User Management, Dashboard, Reporting). **Technology Stack: Frontend:** React.js (with API integration), **Backend:** Spring Boot REST API, **Database:** MySQL with JPA/Hibernate, **Security:** JWT-based authentication, **Deployment:** Docker containers, optional Docker Compose, **Integration:** End-to-end integration between frontend, backend, and database.**Testing:** API testing with Postman; frontend and backend error handling. **Documentation:** Source code, design documentation, API documentation, and user manual. **Presentation:** Live demo, project report, and presentation slides.

3. **Project Deliverables:- Project Proposal:** Problem statement, Objectives, Project scope, Technology stack.**Design Documentation:** System architecture diagram (MVC, API flow), Database schema (ER diagram), UI wireframes/mockups, **Implementation:** Frontend and backend code repositories (GitHub or equivalent), RESTful API endpoints, JWT authentication and role-based access control, Containerized deployment with Docker, **Testing:**Test cases and screenshots (Postman / frontend validation), **Final Report:** Introduction, methodology, results, testing summary, and conclusion, Challenges faced and solutions applied, Future scope. **Presentation:** 8–10 minutes demo, Architecture explanation, Live application demo, Q&A session.

Guidelines for Capstone Mini-Project

- Students shall form a group of 3 to 4 students, and a group shall not be allowed less than three or more than four students, as it is a group activity.
- Students should do the survey and identify needs, which shall be converted into a problem statement for the mini project in consultation with the faculty supervisor/head of department/internal committee of faculty.
- Students' hall submits an implementation plan in the form of a Gantt/PERT/CPM chart, which will cover the weekly activity of a mini-project.
- A log book to be prepared by each group, wherein the group can record weekly work progress, and the guide/supervisor can verify and record notes/comments.
- Faculty supervisor may give inputs to students during the mini project activity; however, focus shall be on self-learning.
- Students in a group shall understand the problem effectively, propose multiple solutions, and select the best possible solution in consultation with the guide/ supervisor.
- Students shall convert the best solution into a working model using various components of their domain areas and demonstrate.
- The solution to be validated with proper justification and a report to be compiled in a standard format of the University of Mumbai.

Guidelines for Assessment of Capstone Mini-Project:

Term Work

- The review/ progress monitoring committee shall be constituted by the head of departments of each institute. The progress of the mini project to be evaluated on continuous basis, minimum two reviews in each semester.
- In continuous assessment, focus shall also be on each individual student, assessment based on the individual's contribution in group activity, their understanding, and response to questions.
- Distribution of Term work marks for both semesters shall be as below;
    - Marks awarded by guide/supervisor based on above practical list and assignment completed by the students : 10
    - Marks awarded by guide/supervisor for Capstone Project Completion: :10
    - Quality of Project report & Project Review : 05

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Review/progress monitoring committee may consider following points for assessment based of the semester project as mentioned in general guidelines.

- In this case in students' group shall complete project in all aspects including,
    - Identification of need/problem
    - Proposed final solution
    - Procurement of components/systems
    - Building prototype and testing
- Two reviews will be conducted for continuous assessment,
    - First shall be for the finalisation of problem and proposed solution
    - Second shall be for the implementation, testing and validation of solution.

Assessment criteria of Capstone Mini-Project.

Capstone Mini-Project shall be assessed based on following criteria;

1. Quality of survey/ need identification
2. Clarity of Problem definition based on need.
3. Innovativeness in solutions/ Novelty in solutions.
4. Feasibility of proposed problem solutions and selection of best solution
5. Collection of Dataset.
6. Cost effectiveness
7. Societal impact
8. Innovativeness

9. Cost effectiveness and Societal impact
10. Full functioning of working model as per stated requirements
11. Effective use of skill sets
12. Effective use of standard engineering norms
13. Contribution of an individual's as member or leader
14. Clarity in written and oral communication

Guidelines for Assessment of Capstone Mini-Project Practical/Oral Examination:

- Report should be prepared as per the guidelines issued by the University of Mumbai.
- Mini Project shall be assessed through a presentation and demonstration of working model by the student project group to a panel of Internal and External Examiners preferably from industry or research organisations having experience of more than five years approved by head of Institution.
- Students shall be motivated to publish a paper based on the work in Conferences/students competitions.

Assessment :

Term Work: Term Work shall consist of list of all practicals' based on the above list. Also, the Term work Journal must include at least 2 assignments and Mini-Project Report.

Term Work Marks: 50 Marks (Total marks) = 10 Marks (Experiment) + 5 Marks (Assignments) + 30 Marks (Capstone Mini- Project with full prototype/ product demo, testing, validation and Report) + 5 Marks (Attendance).

Oral Exam: An Oral exam will be held based on the Capstone Mini-Project.

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